

IAM FOR IDENTITY MANAEGEMENT

Access Control Report

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1. Objectives / Description:

Students will learn how to:

- Create IAM users and groups
- Assign permissions using policies
- Implement least privilege access
- Test access control using IAM credentials

2. Software's / Tools Required:

- Hardware: PC, Internet access
- Software: AWS Free Tier account (Huzaiifa Zahid's account)

3. Lab Environment and Setup:

In this lab we have to setup AWS cloud free tier account on the web.

4. Lab Procedure & Results:

Part 1: Setting up AWS (Free Tier)

After creating the account navigate to the dashboard by typing IAM on the search bar

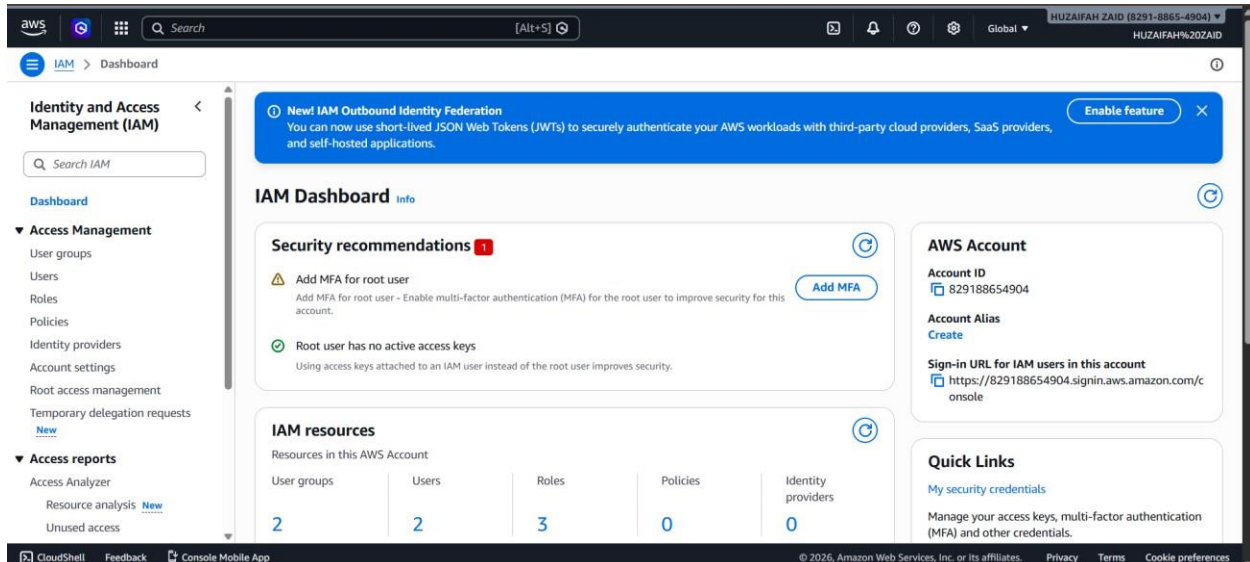
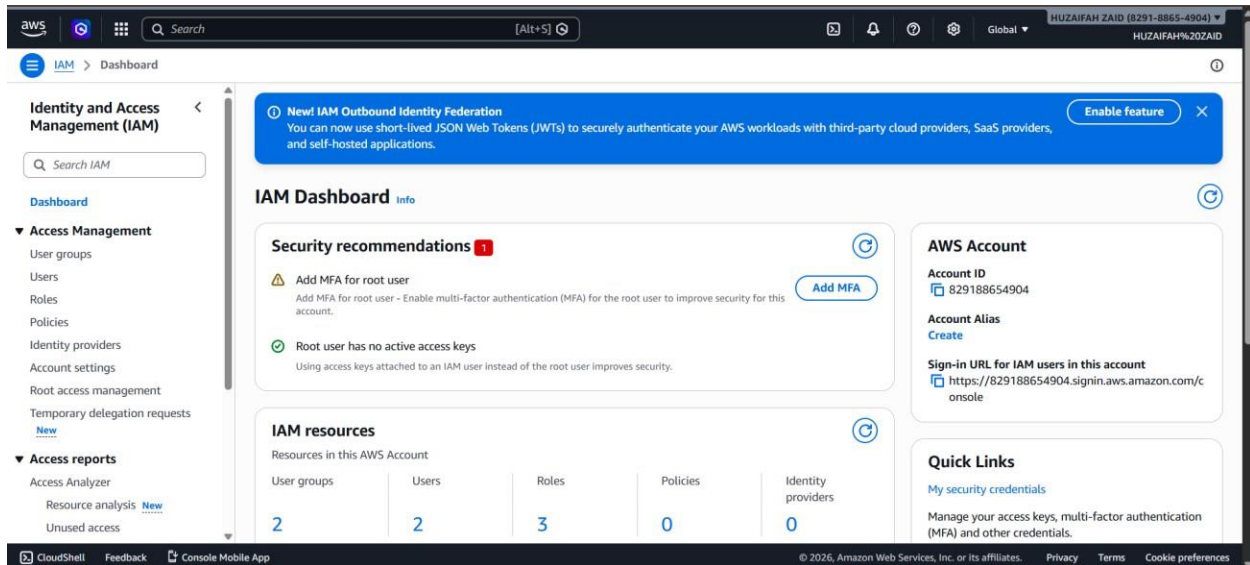


Figure 1

Part 2: Lab Tasks

Task 01 Login to AWS Console:

1. Go to AWS Management Console
2. Login using root account (only for setup)
3. Open IAM Dashboard



Task 02 Create IAM Group:

Navigate to User Groups → Create group

Create two groups:

- Students_Group
- Lab_Assistants_Group

☐	Group name	Users
☐	<u>Hadiya_Group</u>	0
☐	<u>Lab_Assistants_Group</u>	1
☐	<u>Maam.Mahnoor_Group</u>	0
☐	<u>Students_Group</u>	1

Task 03 Attach Policies to Groups:

Attach AWS managed policies:

- For Students_Group:
 - ReadOnlyAccess

group ⓘ

Hadiya_Group [Info](#)

[Delete](#)

Summary [Edit](#)

User group name Hadiya_Group	Creation time March 31, 2026, 11:17 (UTC+05:00)	ARN  arn:aws:iam::829188654904:group/Hadiya_Group
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Users **Permissions** Access Advisor

Permissions policies (0) ⓘ [Refresh](#) [Simulate](#) [Remove](#) [Add permissions](#)

You can attach up to 10 managed policies.

Filter by Type: All types

☐	Policy name ↗	Type	Attached entities
No resources to display			

[Attach policies](#)
[Create inline policy](#)

Attach permission policies to Hadiya_Group

► Current permissions policies (0)

Other permission policies (1/1132) ⓘ

You can attach up to 10 managed policies to this user group. All of the users in this group inherit the attached permissions.

Filter by Type: All types 226 matches

☐	Policy name	Type	Used as	Description
☒	 AiDevOpsAgentReadOnlyAccess	AWS managed	None	-
☐	 AiOpsReadOnlyAccess	AWS managed	None	-
☐	 AlexaForBusinessReadOnlyAccess	AWS managed	None	-

- For Lab_Assistants_Group:
 - AmazonEC2FullAccess

Attach permission policies to Maam.Mahnoor_Group

► Current permissions policies (0)

Other permission policies (1/1132) ⓘ

You can attach up to 10 managed policies to this user group. All of the users in this group inherit the attached permissions.

Filter by Type: All types 1 match

☐	Policy name	Type	Used as	Description
☒	 AmazonEC2FullAccess	AWS managed	Permissions policy (1)	Provides full access to Amazon EC2 via th...

[Cancel](#) [Attach policies](#)

Task 04 Create IAM Users:

1. Go to Users → Create user

2. Create users:

- student1
- assistant1

3. Enable:

o AWS Management Console access

o Set custom password

4. Assign:

o student1 → Students_Group

o assistant1 → Lab_Assistants_Group

STUDENT

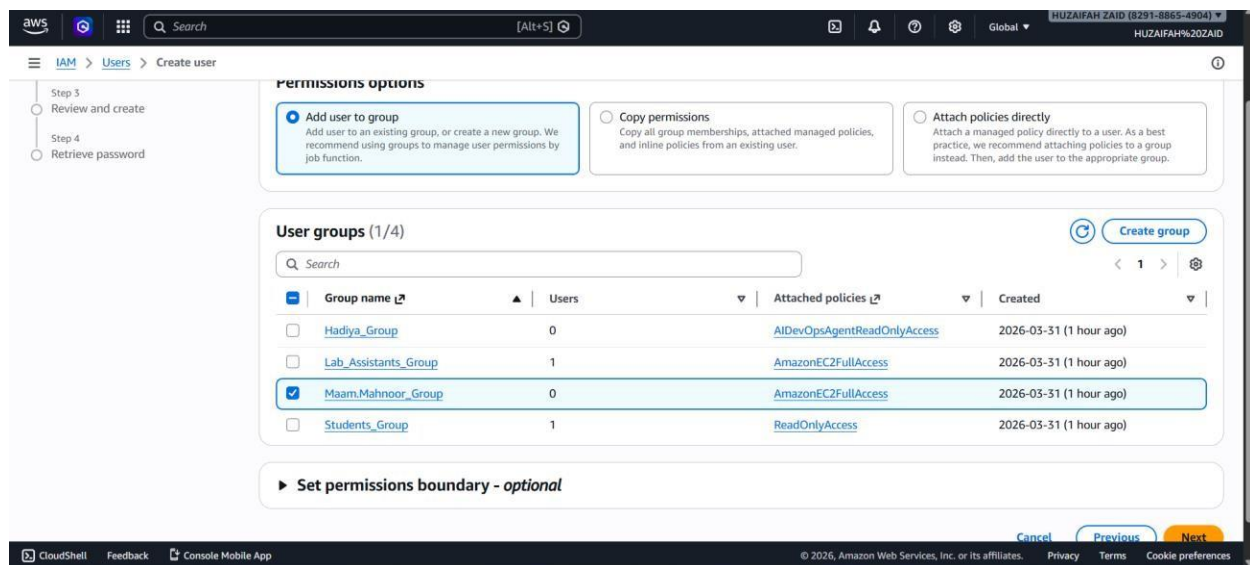
The screenshot shows the AWS IAM console 'Create user' wizard, specifically the 'Set permissions' step. The left sidebar shows the progress: Step 1 (Specify user details), Step 2 (Set permissions), Step 3 (Review and create), and Step 4 (Retrieve password). The main content area is titled 'Set permissions' and includes a sub-header 'Permissions options' with three radio buttons: 'Add user to group' (selected), 'Copy permissions', and 'Attach policies directly'. Below this is a 'User groups (1/4)' section with a search bar and a table of available groups.

Group name	Users	Attached policies	Created
☒ Hadiya_Group	0	AIDevOpsAgentReadOnlyAccess	2026-03-31 (58 minutes ago)
☐ Lab_Assistants_Group	1	AmazonEC2FullAccess	2026-03-31 (1 hour ago)
☐ MaamMahnoor_Group	0	AmazonEC2FullAccess	2026-03-31 (58 minutes ago)
☐ Students_Group	1	ReadOnlyAccess	2026-03-31 (1 hour ago)

In the same way create for Assistant

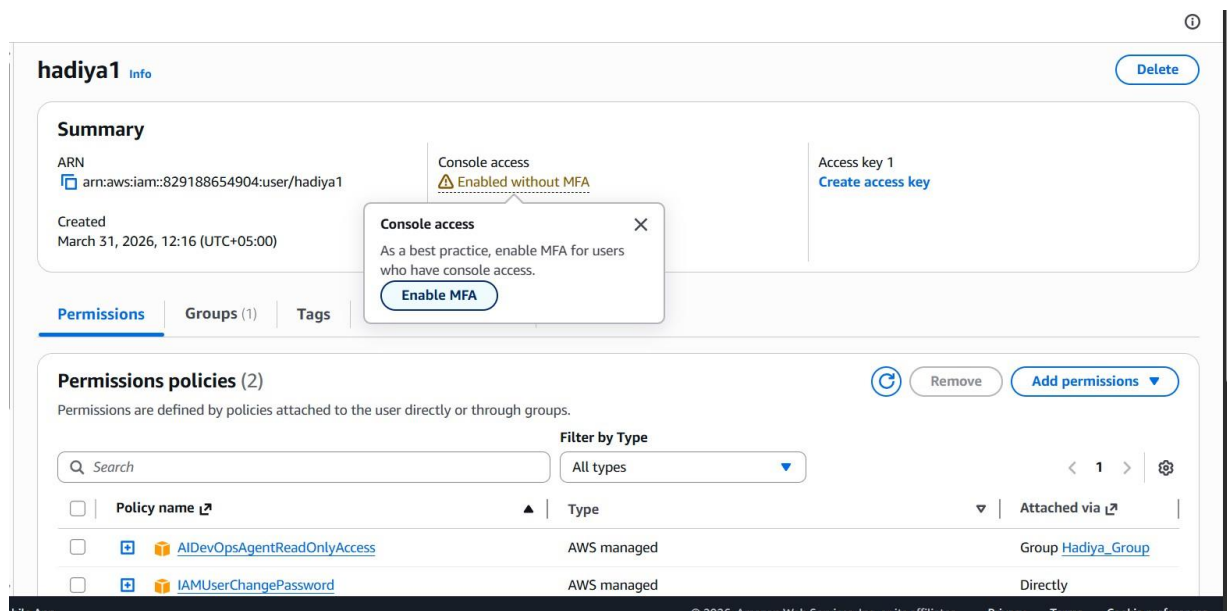
ASSISTANT

Network Security



Task 05 Enable MFA (Multi-Factor Authentication)

1. Go to IAM → Users
2. Select a user
3. Enable MFA using:
 - Google Authenticator / Authy



Network Security

The screenshot shows the 'Assign MFA device' page in the AWS IAM console for user 'hadiya1'. The page has a breadcrumb trail: IAM > Users > hadiya1 > Assign MFA device. On the left, there is a sidebar with 'Set up device' and a plus icon. The main content area is titled 'MFA device name' and contains a text input field with 'hadiyasdevice'. Below this, there is a section titled 'MFA device' with 'Device options'. It lists two options: 'Passkey or security key' (with a fingerprint icon) and 'Authenticator app' (with a smartphone icon). The 'Authenticator app' option is highlighted with a blue border. The footer of the console shows '© 2026, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

Install Google authenticator on your phone, scan QR code and follow the steps

The screenshot shows the 'hadiya1' user page in the AWS IAM console. The breadcrumb trail is IAM > Users > hadiya1. The left sidebar shows 'Identity and Access Management (IAM)' with a search bar and a list of navigation items: Dashboard, Access Management (expanded), Access reports, and Access Analyzer. The 'Access Management' section includes User groups, Users, Roles, Policies, Identity providers, Account settings, Root access management, and Temporary delegation requests. The 'Access reports' section includes Resource analysis and Unused access. The main content area for 'hadiya1' has a 'Delete' button and a 'Summary' section. The 'Summary' section displays the ARN (arn:aws:iam::829188654904:user/hadiya1), Console access (Enabled with MFA), Access key 1 (Create access key), Created date (March 31, 2026, 12:16 (UTC+05:00)), and Last console sign-in (Never). Below the summary are tabs for Permissions, Groups (1), Tags, Security credentials (selected), and Last Accessed. The 'Security credentials' tab shows the 'Console sign-in' section with a 'Console sign-in link' (https://829188654904.signin.aws.amazon.com/console) and a 'Console password' (Updated 11 minutes ago (2026-03-31 12:16 GMT+5)). The 'Last console sign-in' is 'Never'. At the bottom, there is a 'Multi-factor authentication (MFA) (1)' section with 'Remove', 'Resync', and 'Assign MFA device' buttons. The footer of the console shows '© 2026, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.

Do the same for assistant user

Network Security

The screenshot shows the AWS IAM console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and user information for 'HUZAIFAH ZAID (8291-8865-4904)'. The main header indicates the current path is 'IAM > Users > maamMahnoor_Assitant'. On the left, a sidebar menu lists various IAM services under 'Identity and Access Management (IAM)' and 'Access reports'. The main content area features a green notification banner stating 'MFA device assigned' with a close button. Below this, the user 'maamMahnoor_Assitant' is selected, with a 'Delete' button. A 'Summary' section displays the user's ARN, console access status (Enabled with MFA), creation date, and last sign-in. A 'Security credentials' tab is active, showing a 'Console sign-in' section with a sign-in link and a 'Console password' section with a 'Manage console access' button.

Task 05 Apply Password Policy

1. Go to IAM → Account settings

2. Set:

- Minimum length: 8
- Require numbers & symbols
- Password expiration: optional

The screenshot shows the 'Edit password policy' page in the AWS IAM console. The navigation bar and header are consistent with the previous image. The main header shows the path 'IAM > Account Settings > Edit password policy'. The 'Edit password policy' section has an 'info' icon. Under 'Password policy', the 'Custom' option is selected. The 'Password minimum length' is set to 8 characters. Under 'Password strength', the requirements for 'Require at least one uppercase letter', 'Require at least one lowercase letter', and 'Require at least one non-alphanumeric character' are checked. The 'Other requirements' section includes checkboxes for 'Turn on password expiration', 'Password expiration requires administrator reset', 'Allow users to change their own password', and 'Prevent password reuse'.

Task 8: Explain

Q1. IAM used for?

IAM is used to implement the principle of least privilege—granting only the necessary permissions required for a task. Key uses include:

- **User and Group Management:** Creating and managing individual users (people or applications) and organizing them into groups to apply shared permissions.
- **Role-Based Access Control (RBAC):** Creating IAM roles to delegate temporary permissions to trusted entities, such as AWS services (EC2, Lambda), external users, or users from other AWS accounts, eliminating the need for long-term credentials.
- **Fine-Grained Permissions:** Using JSON policy documents to define granular rules, such as allowing a user to read data from a specific S3 bucket but not delete it.
- **Multi-Factor Authentication (MFA):** Enforcing an extra layer of protection for users, requiring a password and a one-time code.
- **Credential Management:** Generating and rotating security credentials (passwords, access keys, SSH keys).
- **Access Analysis:** Using IAM Access Analyzer to scan for resources shared with external entities and to generate policies based on actual access activity.

Q2. What are other security features provided by AWS?

Beyond IAM, AWS offers a comprehensive suite of security services for infrastructure, data, and application protection:

- **Identity & Access:** AWS IAM Identity Center for centralized access management, and Amazon Cognito for user sign-in in applications.
- **Detection & Monitoring:** Amazon GuardDuty for threat detection, AWS CloudTrail for activity logging, and AWS Security Hub for centralized compliance and security findings. Amazon Inspector and Detective automate vulnerability scanning and investigation.
- **Data Protection & Encryption:** AWS Key Management Service (KMS) for encryption keys, Amazon Macie for protecting sensitive data with ML, and AWS Secrets Manager for secret rotation.
- **Infrastructure & Network Protection:** AWS Shield for DDoS protection, AWS WAF for web application security, and Amazon VPC for network isolation.
- **Governance & Compliance:** AWS Organizations for account management and SCPs, and AWS Artifact for compliance reports.